

Functional Performance Specifications

a data collection system designed and assembled for the purpose of capturing all aspects of human motion.

1. System that supports real time data collection from Ascension magnetic trackers, Polhemus' magnetic trackers, NDI's Optotrak Certus, Natural Point's Optitrack cameras, Qualysis' cameras, Motion Analysis Corp's cameras, Vicon cameras, PhaseSpace, Metria, Xsens Awinda IMUs and APDM IMUs along with Bertec, AMTI or Kistler force plates; Bertec and AMTI Instrumented Treadmills; Noraxon, Delsys, or any analog EMG systems; ATI 6DoF transducers; SR Research's EyeLink II or Tobii Pro eye tracking system; BioSemi, ANTNeuro and MindMedia EEG systems; VR displays, haptic devices, event markers, and high speed digital videos. The benefit of this capability is reduced cost as research evolves and additional hardware is required. In addition operating costs are lower as a common software interface is used for all hardware.
2. Data is collected in real-time at the frame rate of the hardware (data is not down-sampled).
3. Multiple kinematic technologies can be combined into a hybrid motion capture system to simultaneously take advantage of the strength of each technology for tracking and animating a single subject, i.e. use a camera system simultaneously with electromagnetic trackers or IMUs.
4. Data and active animation windows displaying raw and/or processed are available for viewing in both live streaming mode and during recording for quality check of data.
5. Data can be collected from one, all or any combination of supported hardware and are fully integrated and synchronized.
6. Data are collected, processed and immediately available for viewing with scaled and proportioned animation models.
7. Data can be easily displayed in multiple ways such as time-series graphs, bar graphs, x-y plots or text in an animation window.
8. Finger movements can be tracked and analyzed with each individual bone instrumented with a 6-DoF tracker to animate and provide data from each phalange. Or, finger movements can be tracked and analyzed with only the distal phalanx instrumented with a 6-DoF tracker to interpolate, animate and provide data for the phalanges which are instrumented and not instrumented. This allows evaluation of finger and hand motion with minimal hardware
9. 3-D Objects can be tracked and visualized in an animation window
10. Data, including all standard kinematic and kinetic data, joint forces and moments, helical axes, nonlinear analysis, etc., are selectable from intuitive drop down menus eliminating the need for programming capability on the part of the user. In addition, users can define their own data using standard math notation which are then viewable in real time.
11. Real time biofeedback functionality maps any raw or processed data into 1, 2 or 3 dimensional space. Feedback can be in the form of any combination of visual images, audible signals or digital to analog voltage output. This functionality provides i) a method to ensure subjects perform tasks consistently trial to trial during rehabilitation ii) a basis for proprioceptive studies, iii) a method to track reaction times in response to various stimuli, and iv) a way to drive robots or other devices.
12. Co-location routines ensure Virtual Reality displays, haptic devices, eye trackers, force plates, and hand transducers report data in a common reference frame. This eliminates the need to post-process data and ensures that data can be reported in real time.
13. Dynamic control of the belt speed and acceleration of the Bertec Instrumented Treadmill using user-defined formulas for self-paced walking, introducing perturbations, balance studies, etc.

14. CT/MRI registration functions provide a means to generate actual bone geometry and joint interaction in an animation window. Data collected pre and post intervention can then be viewed for changes in joint performance.
15. Real-time eye-tracking for 3D gaze with head compensation using any of the supported 6-DoF kinematic trackers.
16. Icons and scripts are used to create a unique interface and workflow for user-defined setup, functions, analysis, data displays, images and messages. This results in an easy-to-use, efficient and robust interface, reducing user error during operation.